When to Use Topical vs. Oral Therapy in Acne

As shown in Fig. 17, **topical therapy** is first-line for patients with **mild-to-moderate acne**. Several factors influence the decision to initiate **systemic therapy**, including:

- Degree of inflammation
- Truncal involvement
- Duration of acne
- Decreased quality of life
- Presence or risk of scarring

Early, effective treatment is crucial, as the risk of scarring increases with delayed intervention. Topical therapy is also ideal for **maintenance treatment** and should include a **retinoid** with or without **benzoyl peroxide (BPO)**, or **azelaic acid**—particularly in patients who are pregnant or planning pregnancy.

Systemic Acne Treatment

Evidence-based systemic treatments for acne include:

- **Oral tetracyclines** (used in combination with topical agents such as retinoids, azelaic acid, and BPO)
- **Oral isotretinoin**
- **Anti-androgenic pills**

Zinc may benefit patients with confirmed deficiency. However, excessive intake of zinc (and selenium) can lead to adverse effects. Zinc supplementation may be considered during pregnancy as a supportive measure.

Biologics and **non-steroidal anti-inflammatory drugs (NSAIDs)** lack strong evidence for efficacy in acne, although specific agents like **zileuton** have shown effectiveness in some studies.

Oral Antibiotics

Antibiotics have long been a cornerstone of acne management. However, growing concerns about **antibiotic resistance** and national and international efforts to promote **antibiotic stewardship** have led experts to advocate for **antibiotic-sparing strategies**.

Effective oral antibiotics for acne include:

- **Tetracyclines**: tetracycline, doxycycline, minocycline, lymecycline
- **Trimethoprim ± sulfamethoxazole**
- **Macrolides**: erythromycin, azithromycin
- **Amoxicillin**, **cephalexin**

Tetracyclines are first-line for **moderate-to-severe acne**, unless contraindicated. **Macrolides** are an alternative when tetracyclines are unsuitable. **Penicillins and cephalosporins** have limited supporting data but may be used in pregnancy or in patients with allergies to other antibiotic classes.

Oral antibiotics primarily exert **anti-inflammatory effects**, with secondary activity against *Cutibacterium acnes*. They should be prescribed for **no longer than 3 months** to minimize the risk of microbial resistance. Prolonged use should be avoided.

According to the **American Academy of Dermatology (AAD) guidelines**, oral antibiotics are indicated for **moderate-to-severe inflammatory acne** and **must be combined** with **topical retinoids and BPO** to enhance efficacy and reduce resistance.

Prescribing practices in the **Middle East** are largely based on clinical judgment due to limited regional data. Additionally, there is a lack of structured **antibiotic stewardship** and educational programs in dermatology.

Hormonal Therapy

Hormonal therapy can be an effective option for **women with acne**, particularly those with hormonal imbalances or signs of hyperandrogenism. However, cultural and societal norms in the Middle East may create barriers to discussing or prescribing **estrogen-containing combined oral contraceptives (COCs)**, especially for unmarried or young women.

Dermatologists are encouraged to become familiar with available hormonal options and consider **collaborating with gynecologists** when appropriate.

Hormonal therapy should **not be used as monotherapy**, as its onset of action is slow. Initial improvements typically include:

- Reduction in **comedones**
- Decreased **seborrhea**
- Improvement in **inflammatory lesions** after 8–10 weeks

Both the **European Dermatology Forum (EDF)** and **AAD guidelines** recommend hormonal therapy as an adjunctive treatment for women with acne.

Table 10: American Academy of Dermatology Recommendations for Oral Antibiotics in Acne

Systemic antibiotics are recommended for:

- **Moderate and severe acne**
- **Inflammatory acne resistant to topical therapy**

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